

NumStability in AI-Assisted Lean Formalization and Proof

A reproducible, exploratory ten-task N/L benchmark report

HighamBench evidence report

28 September 2026

Abstract

We compare a Mathlib-only condition (N) with the same Lean environment plus a frozen NumStability library and one task-neutral orientation conversation (L). All ten reported paper tasks ended with an audited faithful statement and a zero-hole, kernel-accepted proof in both conditions. Across these pairs, L used 19.7% fewer nonblank proof-code lines (2,537 versus 3,158) and 22.5% less contestant-active time (5,410 versus 6,979 seconds), while using 34.7% more net-new tokens (1.716 versus 1.273 million). Charging the one-time L scout once leaves a 19.9% time advantage and 41.7% token excess. The formalization stage was 7.3% slower for L; the proof stage was 29.9% faster. At realized-reuse scores 1, 2, and 3, aggregate time gains were -9.1% , $+19.3\%$, and $+31.1\%$. These are descriptive development results, not a held-out causal estimate: task selection was outcome-informed, five tasks had previously favorable results, and the reuse rubric was annotated after outcomes. The five newly screened tasks alone used 6.4% more L proof lines. ^[1, 4, 23]

1 Question, corpus, and scientific status

The question is whether a numerical-stability library reduces the work and Lean code needed for an AI formalizer to state *and prove* selected paper results. Both conditions receive the same hashed source paper, task packet, Lean 4, Mathlib, formalization prompt, audit policy, and proof requirement. L alone receives frozen NumStability source and compiled declarations plus an orientation appendix. Neither condition receives a shared Lean theorem scaffold encoding the desired result.^[8, 10, 2]

The ten analyzed tasks span five source clusters: three Castaldo–Whaley–Chronopoulos dot-product tasks, four Blanchard–Higham–Higham log-sum-exp/softmax tasks, and one each from Hallman–Ipsen, Higham, and Rump. Four tasks were marked hard in the frozen task metadata. The corpus is not a random sample of numerical stability: tasks were selected for expected library relevance, and the final reported ten were chosen after outcome review. Five were previously favorable both-proved tasks. This is therefore an *outcome-informed exploratory development corpus*, not a prospective or held-out evaluation. The dated selection manifest is public, while the figures and numeric discussion here use only the ten reported tasks. ^[4, 6, 2]

The measured result combines sealed pairs from two campaign identities. One audit-schema incident required a new, separately identified pair; the valid sealed pairs were not rerun. A pre-contestant deployment-index incident is preserved separately and is not a measured task outcome. The public pair archives retain prompts, candidate hashes, audit decisions, accepted Lean files, proof checks, and hardware records. The ten-task derived ledger preserves original pair identities and source SHA-256 values.^[2, 15, 16]

Cluster	Source	Tasks	PDF SHA-256 prefix
CAST08	Castaldo et al., superblock dot product	3	ffdea8e2136c7f42
H20	Higham, <i>Accuracy and Stability</i> , Problem 20.8	1	7ca85d5caf3ffd5a
HI21	Hallman–Ipsen, summation-tree bounds	1	cce25923a86d5d20
P14	Blanchard et al., log-sum-exp/softmax	4	7247047bc49218e0
RUMP12	Rump, summation/dot-product error	1	4b23ffc2f7588ddc

Table 1: Five paper/source clusters. Selected task-packet and PDF hashes are in `generated/source_registry.json`.^[5, 7]

2 Frozen protocol and execution

2.1 Two stages, two clocks, and an independent audit

A contestant first submits a Lean theorem with a single allowed `sorry` occupying its target proof. There are at most four frozen statement submissions. Each is hashed before off-clock compilation, integrity validation, and independent candidate-specific faithfulness audit. An unfaithful statement receives condition-neutral feedback in the same formalizer conversation. Once a faithful statement is frozen, the model gets up to four submissions to prove that *exact proposition*. A proof with `sorry`, `admit`, changed dependencies, extra axioms, or a trust escape is not accepted. Proof success requires zero-hole Lean validation. The formalizer is explicitly told not to submit a partial proof as complete. [8, 10, 2]

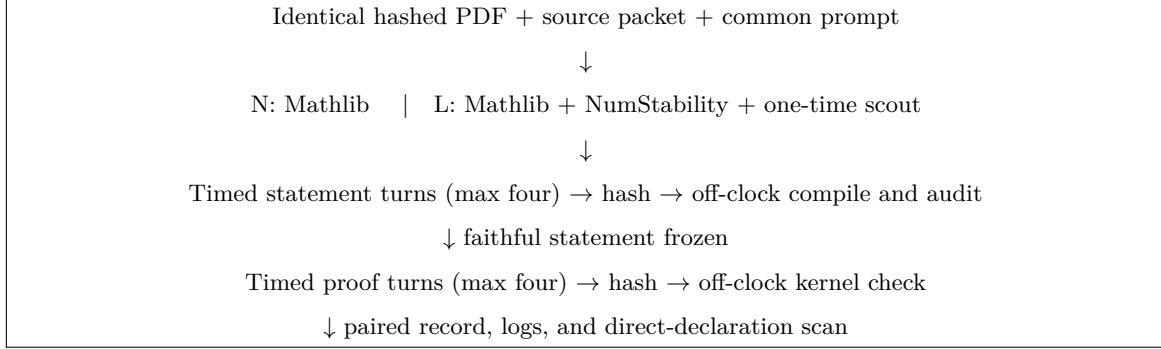


Figure 1: Source-first workflow. Audit and validation cost is recorded but not charged to contestant-active clocks.[8, 9]

Faithfulness is binary. A full-domain equivalent statement or accepted strengthening may pass; a narrowing of the paper’s domain does not. The audit uses fresh, stateless and condition-blind blind-translation, direct, round-trip, and adjudication roles with dependency-aware semantic material. These agents are independent of the formalizers. A faithful statement is not counted as a proof; the zero-hole proof gate is separate. Some accepted statements are equivalent while others are strengthenings, which can affect the proof effort even when both pass.[8, 12, 2]

The formalizer was GPT-6 Sol at high reasoning effort in both conditions; GPT-6 Astra/high performed auditing. Contestant subagents and multi-agent tools were disabled and attested by session controls. Observable tool and turn events were archived; hidden chain-of-thought was not available. Access used ChatGPT sign-in through Codex CLI 0.156.0, not metered API billing.[9, 15, 17]

2.2 Prompts and library access

The common prompt says the PDF is authoritative and asks for the paper’s full domain, quantifiers, operation order, rounding model, constants, and conclusion. It warns of independent faithfulness review and later proof. N receives this prompt with Mathlib. L receives the same prompt plus a short appendix encouraging deliberate discovery of the relevant algorithm representation, compatible floating-point model, and error/perturbation lemmas. L has access to the full library, not an oracle or task-specific retrieval packet. A separate strict no-hole proof prompt is frozen.[10, 11]

Prompt	SHA-256 prefix
Common statement	c3ece6dddf2b2c
L treatment appendix	579a68943cfb636
Task-neutral scout	f2517b44f0801c0
Proof-stage base	b21295790a3ff3f
Strict no-hole proof prompt	537ebad7810ec9a
Audit roles	exact hashes in archived campaign inputs

Table 2: Prompt identity; source files and observed effective prompts in the archives provide full text.[10, 11, 15]

The task-neutral L scout ran once for 183.313 seconds using GPT-6 Sol/high. It used 1,274,811 input tokens, 1,193,216 of them cached, plus 6,873 output tokens: 88,468 net-new tokens by the benchmark formula. The provider total including cached input was 1,281,684 tokens. Its frozen orientation root was forked into task conversations; the scout was never rerun per task. [13, 14]

2.3 Hardware, parallelism, and software

Titan ran Linux/x86-64 on an Intel Core i9-13900K. Three disjoint lanes ran concurrently, each capped at eight logical CPUs, 24 GiB RAM, zero swap, and 512 processes. N/L order alternated by frozen schedule index. CPU affinity, cgroups, before/after machine snapshots, and sampled per-turn peaks were logged. In the ten reported tasks, sampled per-condition maxima were 5.39 cores and 2.35 GiB for N, and 4.70 cores and 2.01 GiB for L. Brief spikes could be missed. Parallelism reduced campaign makespan; it did not divide individual contestant-active time.^[1, 15]

Lean was `leanprover/lean4:v4.29.0-rc3`, Mathlib commit `e8ea1afc32790ce1d4e1a4e45cc412ba9388716b`, and NumStability commit `45813a95dacf577461bae13f033af0dbc985a225`. The separately recorded full-library build took 1,315.18 seconds after Mathlib dependency-cache preparation; it is not charged to task-active time. The build record identifies hardware, cache state, command, output digest, and resource measurements. Authentication material and compiled caches are not redistributed.^[18, 19, 20]

3 Metrics and denominators

Formalization time includes contestant-active work through the frozen faithful statement, including ordinary library search and repairs. Proof time starts after that statement is frozen and includes proof turns and rejected submissions. Their sum is task-active time. Compilation, dossier construction, and auditing are separately measured excluded overhead. The headline time is neither whole campaign makespan nor end-to-end wall time. Net-new tokens equal input minus cached input plus output; full provider usage including cached input is retained separately. Code lines exclude blank and comment-only lines. All ten pairs have faithful statements and complete proofs in both conditions, so paired proof denominators are ten. Warm-inclusive per-task panels allocate the one-time L scout evenly over ten for visualization; aggregate and cumulative calculations charge it once. Positive N-to-L gain means less L cost.^[9, 2, 13]

Metric (ten tasks, summed)	N	L	L/N	L gain
Formalization seconds	1,389.1	1,490.2	1.073	-7.3%
Proof seconds	5,589.8	3,920.0	0.701	+29.9%
Total active seconds	6,978.9	5,410.2	0.775	+22.5%
Total + warm seconds	6,978.9	5,593.5	0.801	+19.9%
Formalization net-new tokens	399,748	980,733	2.453	-145.3%
Proof net-new tokens	873,571	734,899	0.841	+15.9%
Total net-new tokens	1,273,319	1,715,632	1.347	-34.7%
Total + warm net-new tokens	1,273,319	1,804,100	1.417	-41.7%
Statement code lines	333	337	1.012	-1.2%
Proof code lines	3,158	2,537	0.803	+19.7%

Table 3: Machine-generated aggregates. The one-time scout is charged once in the two warm-inclusive rows.^[1, 3]

4 Time, token, and code outcomes

L was faster on eight of ten tasks and used fewer proof-code lines on six of ten. Formalization cost 7.3% more active time and 2.45 times as many net-new tokens for L; proof used 29.9% less time and 15.9% fewer net-new tokens. Mean time was 697.9 seconds/task for N and 541.0 for L, or 559.3 with the scout allocated. Mean net-new tokens were 127.3 thousand for N and 171.6 thousand for L, or 180.4 thousand with the scout. The time and proof-line gains do not imply token savings. Every task label in the figures carries its realized-reuse score as $(R=x/3)$; aggregate plots print the score distribution.^[1, 23, 3]

The one-time cost is best read cumulatively: it is paid before any task, not as a repeated delay. Figure 10 shows that the active-time advantage survives charging it once. The token deficit grows. Cached-prefix processing is reported separately rather than being misrepresented as zero resource use.^[1, 13]

5 Declaration uptake and overlap

The declaration scan uses elaborated constants in final L statements and kernel-accepted proof terms, including proof-local helper bodies. It is not an import or text-match count. In the ten reported tasks, nine had substantive NumStability statement reach, and all ten had direct proof-term reach. Exact names, counts, and statement/proof surfaces are in the machine-readable declaration and reuse ledgers.^[21, 23, 2]

The five score-gain heatmaps share one interpretation: the horizontal axis is realized reuse score, the vertical axis is the named N-to-L gain in percent, and darker cells contain more tasks. An occupied cell prints the exact gain values of its tasks, even when two fall in the same fixed-width bin. The empty score-zero column is shown rather than suppressed. Positive gain means L used less time, fewer tokens, or fewer proof lines. These are descriptive, binned views of ten observations, not fitted relationships. Their exact bin edges and task memberships are published.^[24, 23]

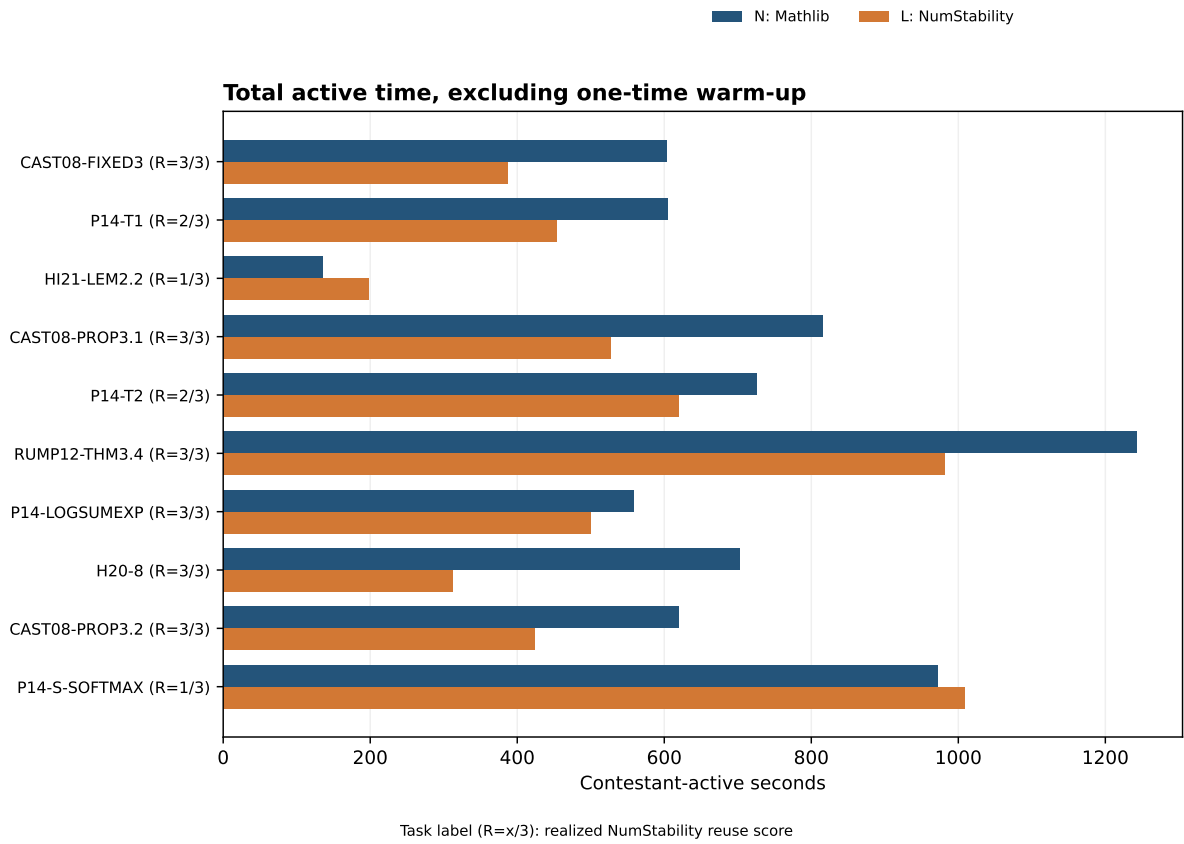


Figure 2: Per-task total contestant-active time, excluding the one-time scout. Task-time library search remains included. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

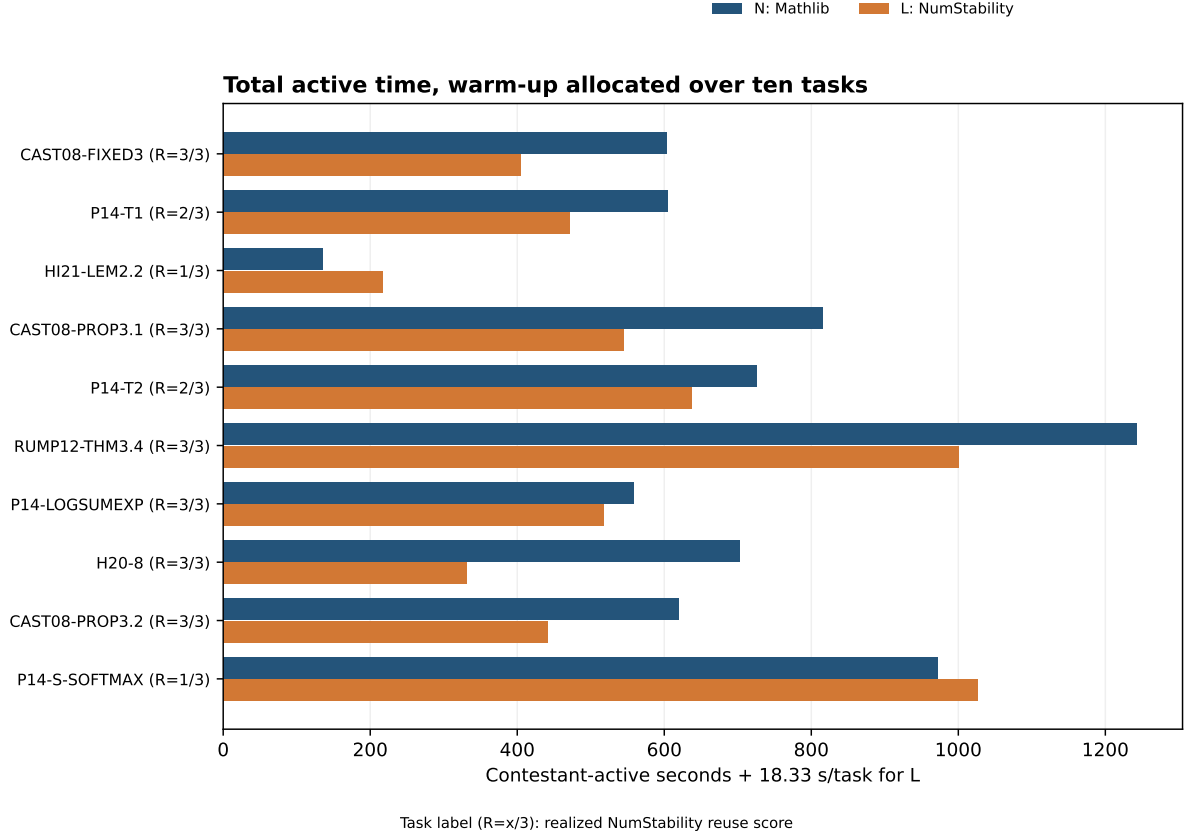


Figure 3: Per-task active time with $183.313/10 = 18.331$ warm seconds allocated to each L bar. This is one scout, not ten. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

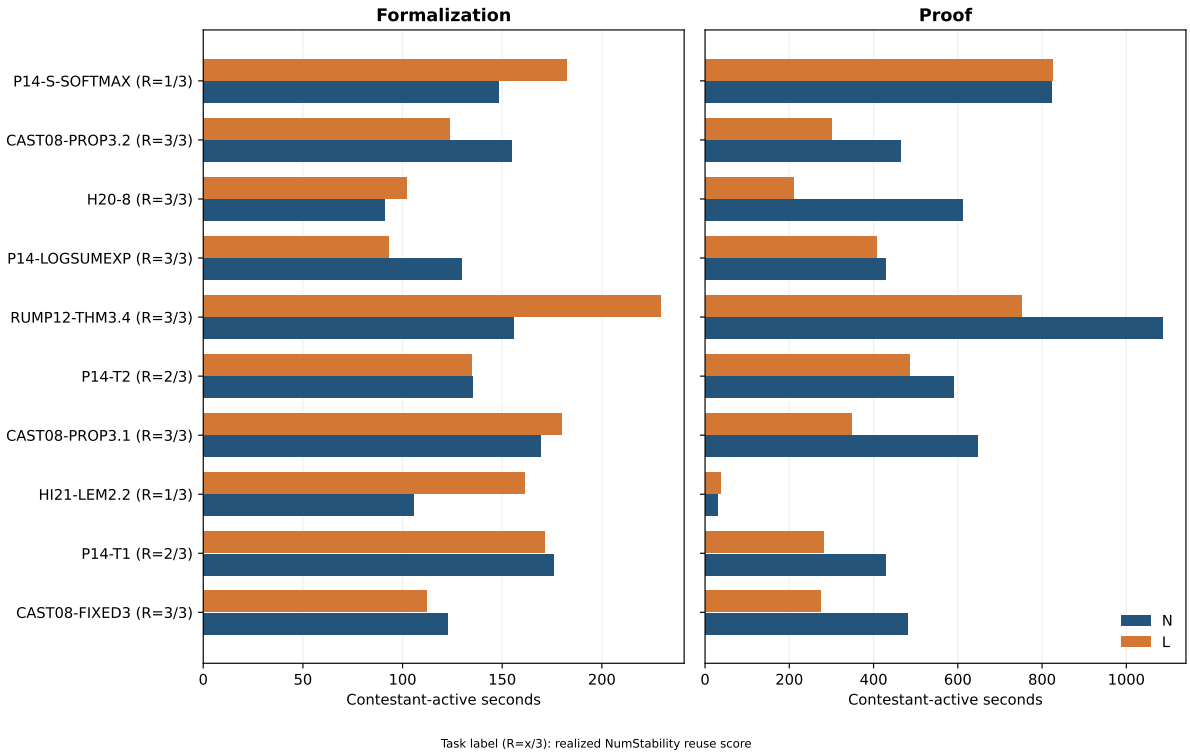


Figure 4: Formalization and proof components of task-active time, excluding off-clock audit and validation. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

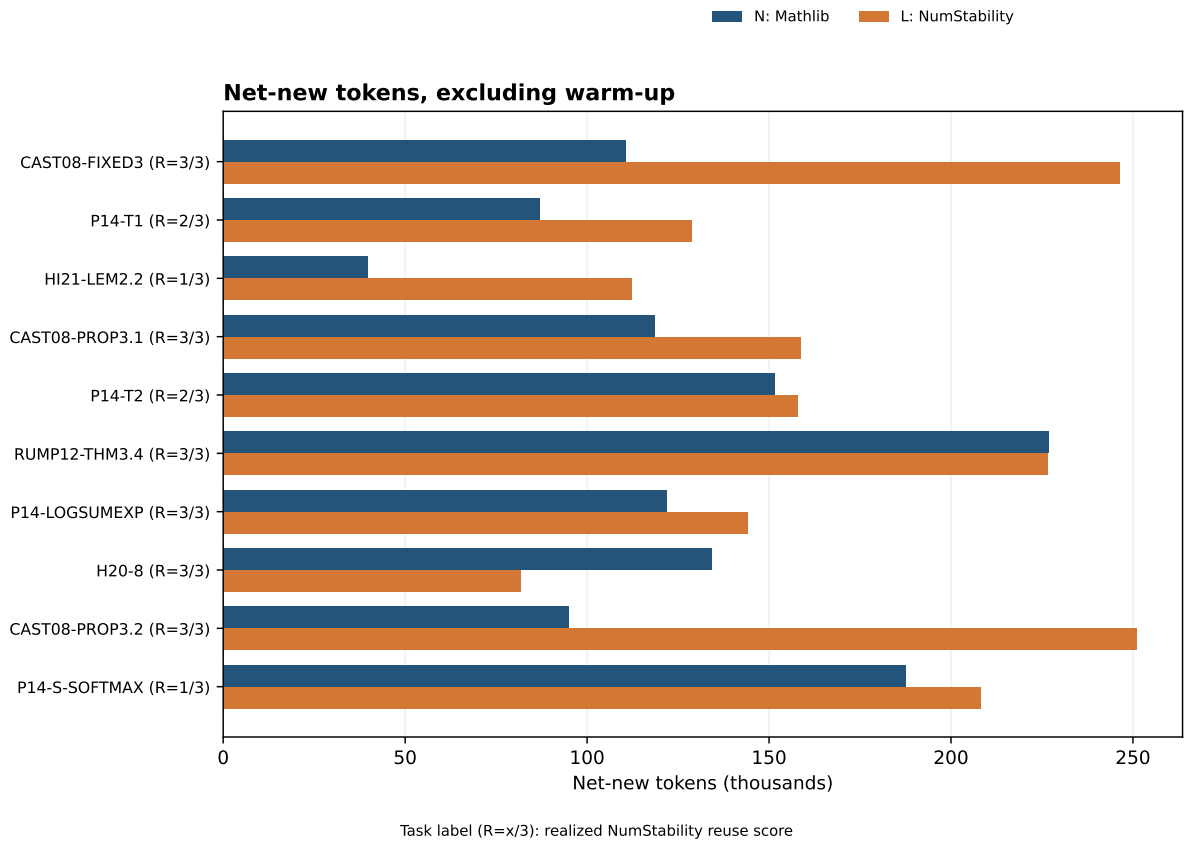


Figure 5: Per-task net-new contestant tokens, excluding the warm scout and cached input. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.^[2, 3]

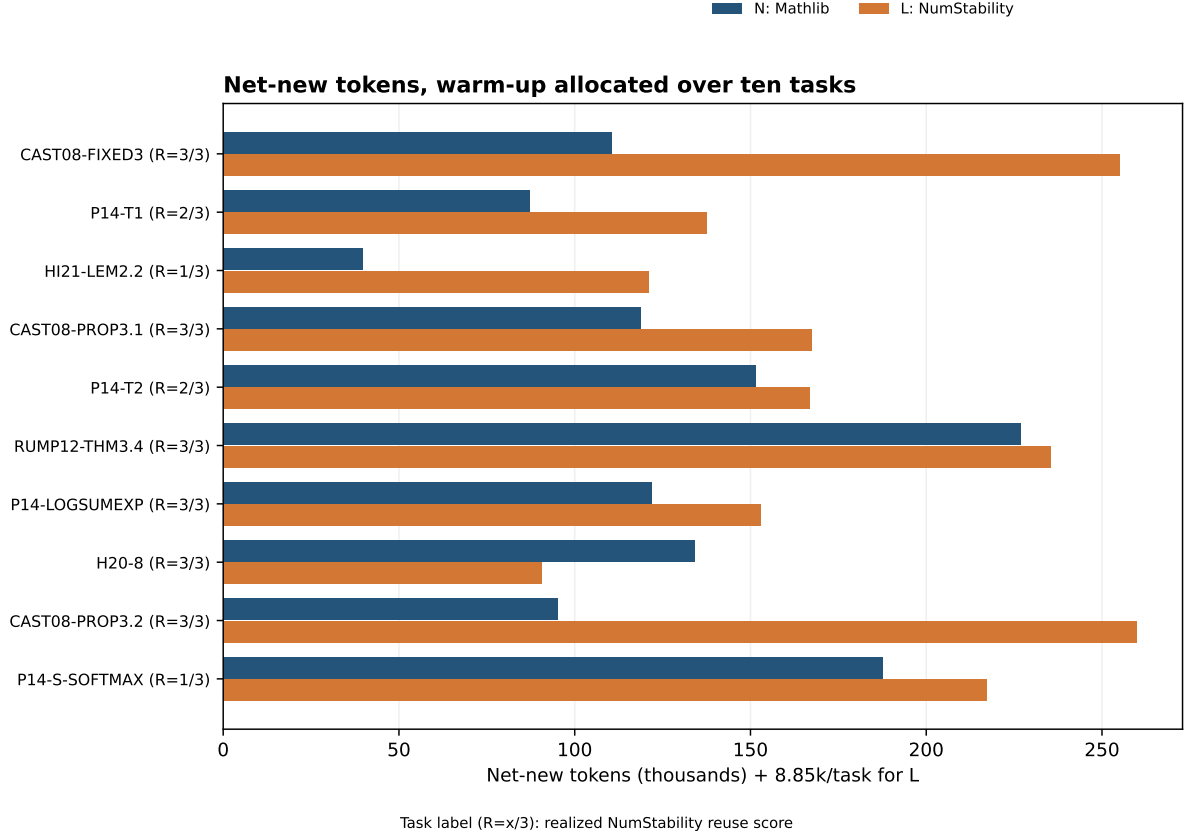


Figure 6: Per-task net-new tokens with $88,468/10 = 8,847$ scout tokens allocated to each L task. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

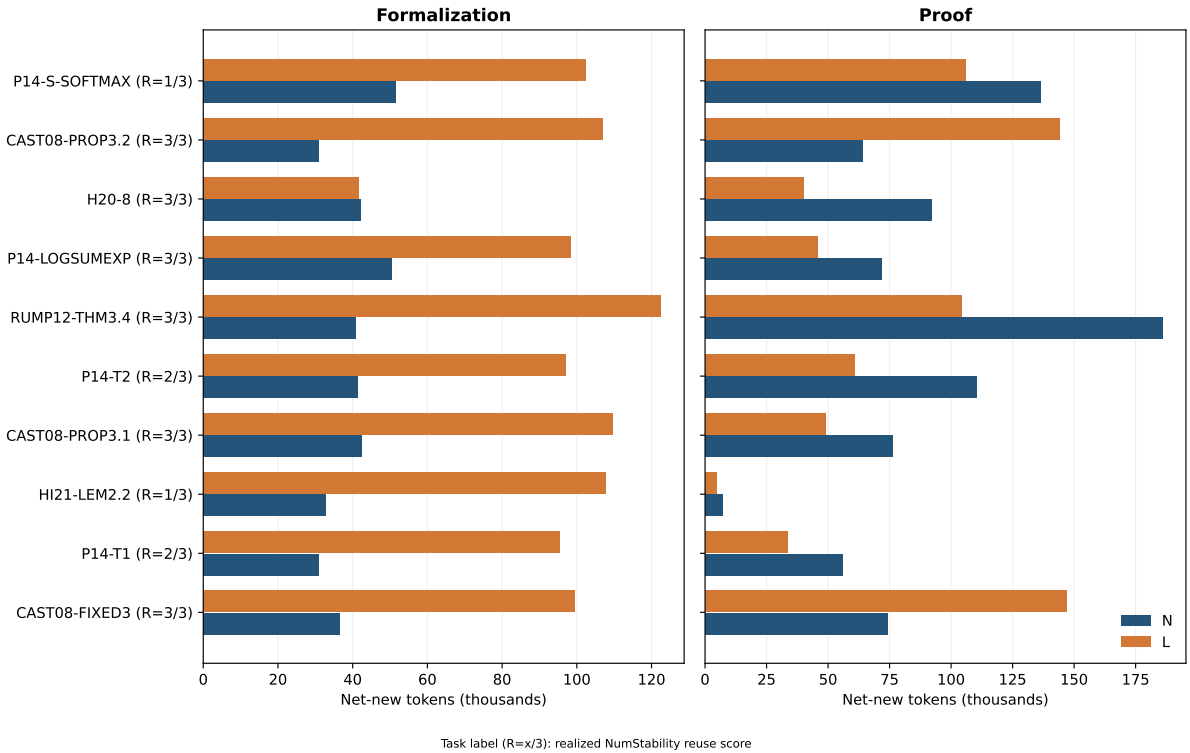


Figure 7: Formalization and proof net-new tokens; L's statement-stage cost is visible separately. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

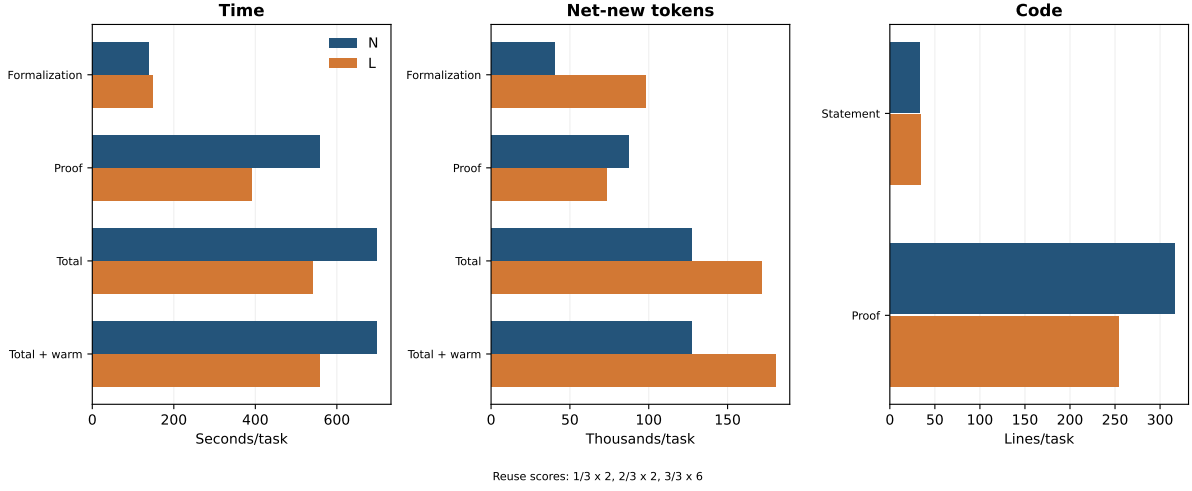


Figure 8: Arithmetic means per task for time, tokens, and code; warm-inclusive bars allocate the scout over ten tasks. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

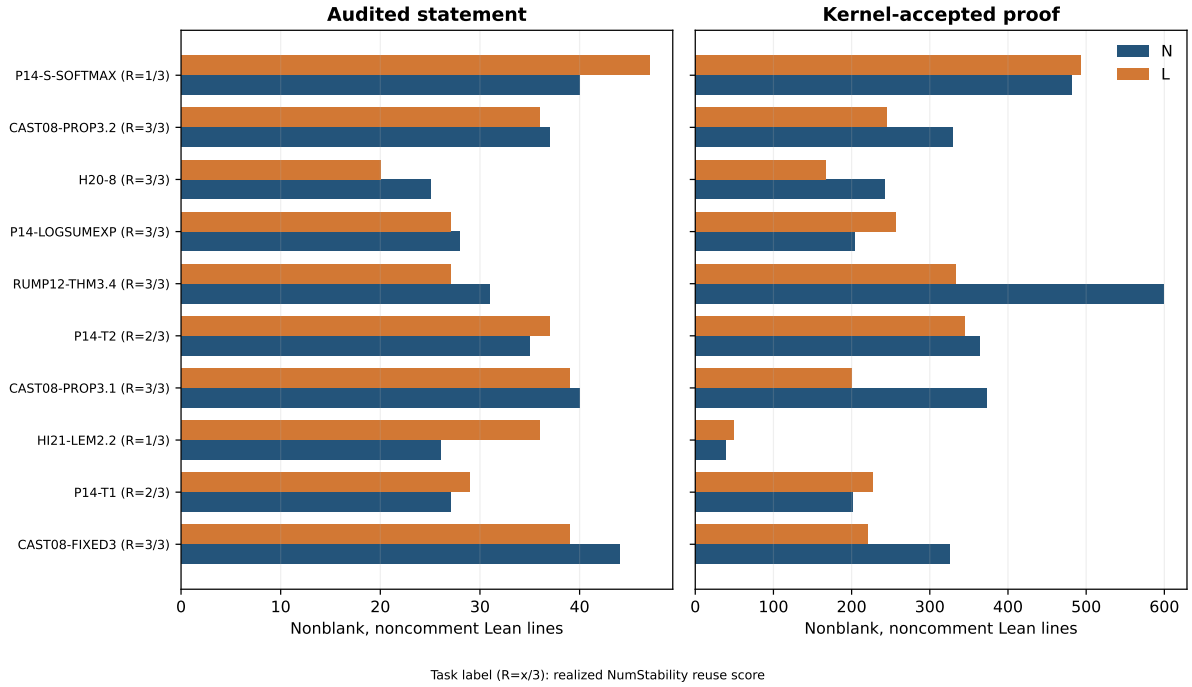


Figure 9: Final faithful statement and complete proof code lines, excluding blanks and comments. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

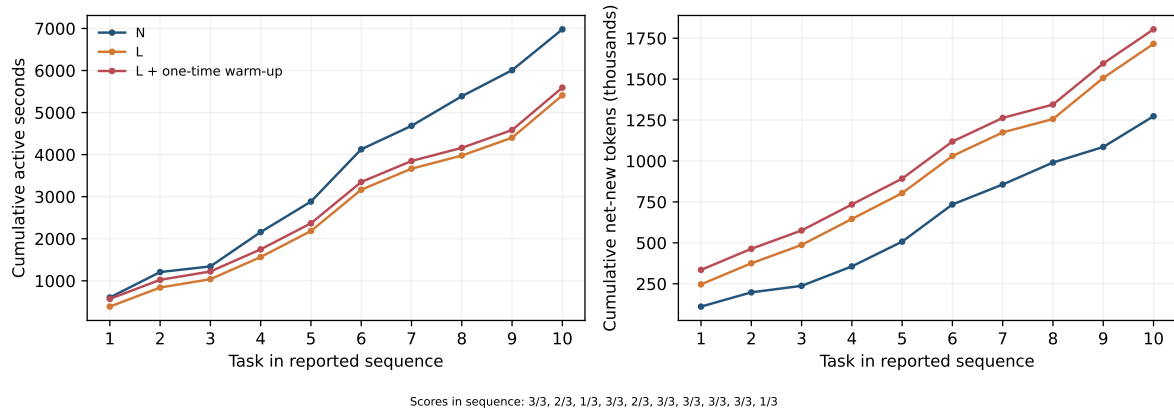
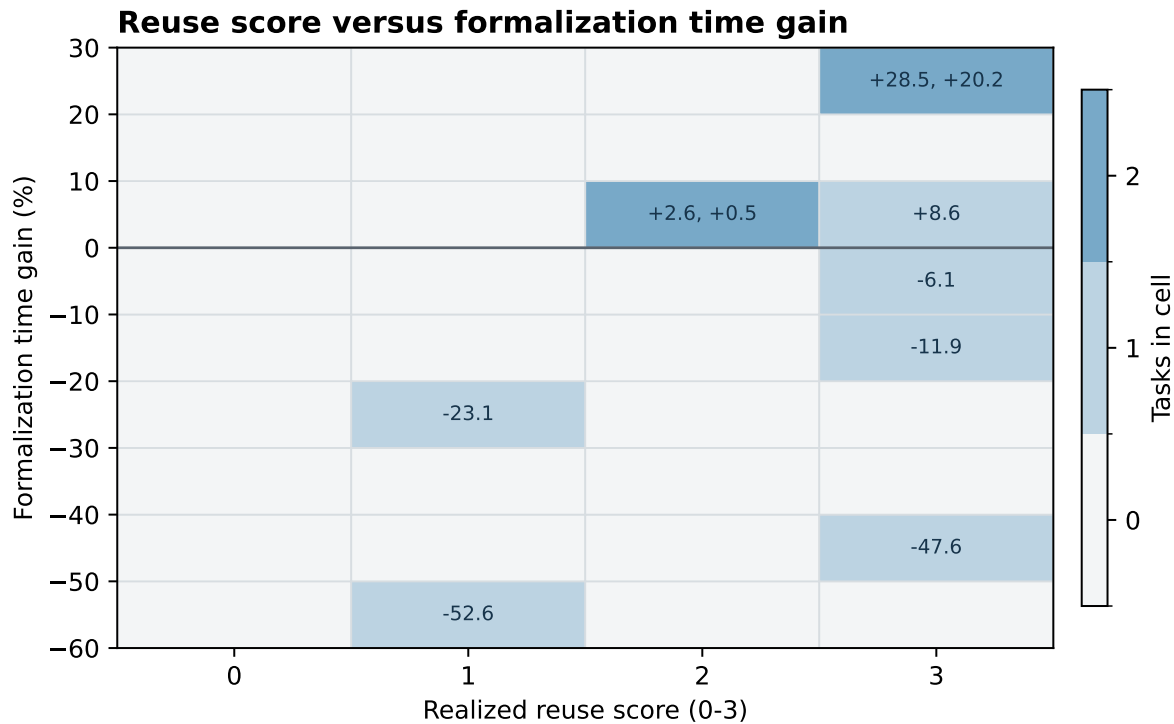


Figure 10: Cumulative contestant-active work in the ten-task order shown. The red L curve charges the one-time scout once. This is not parallel campaign makespan. Each score appears in the footer in plotted order.[1, 3]



Positive gain = L improves on N. Cell labels give the exact task gain(s).

Figure 11: Reuse score versus formalization-time gain. Each occupied cell prints exact task gains in percent; shade encodes the number of tasks. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

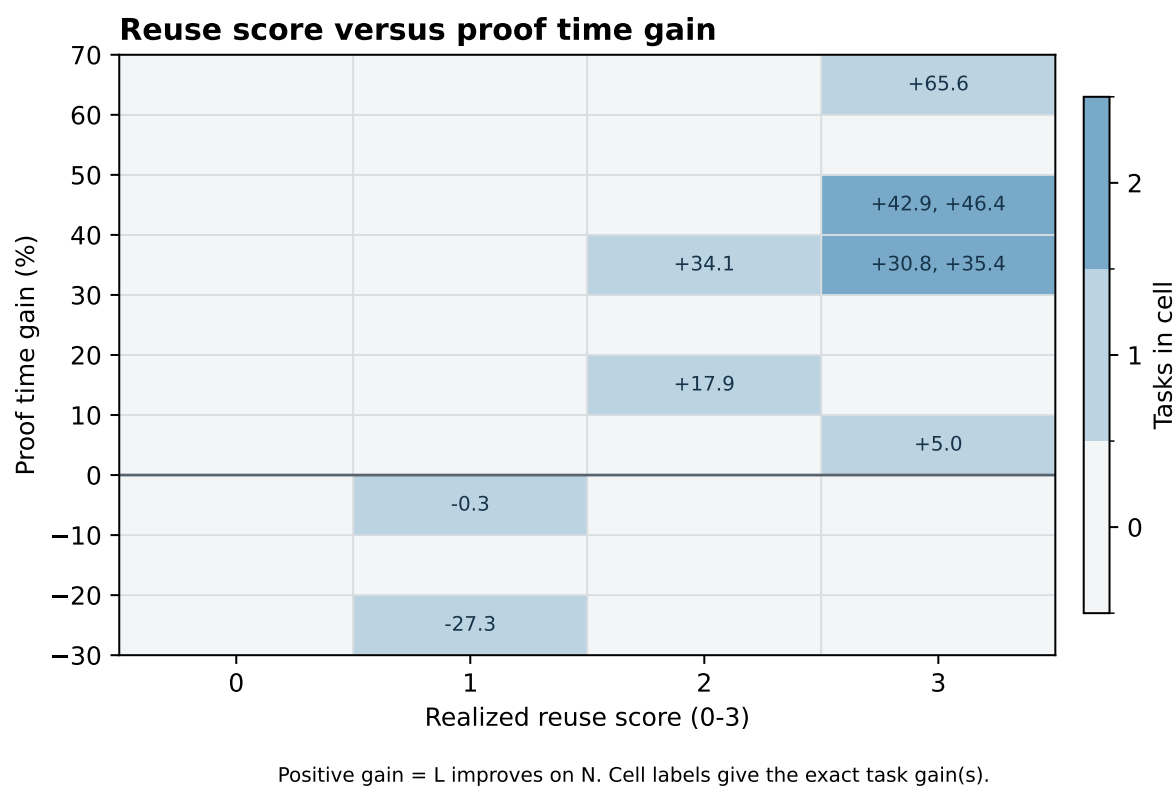


Figure 12: Reuse score versus proof-time gain, with the same cell convention. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

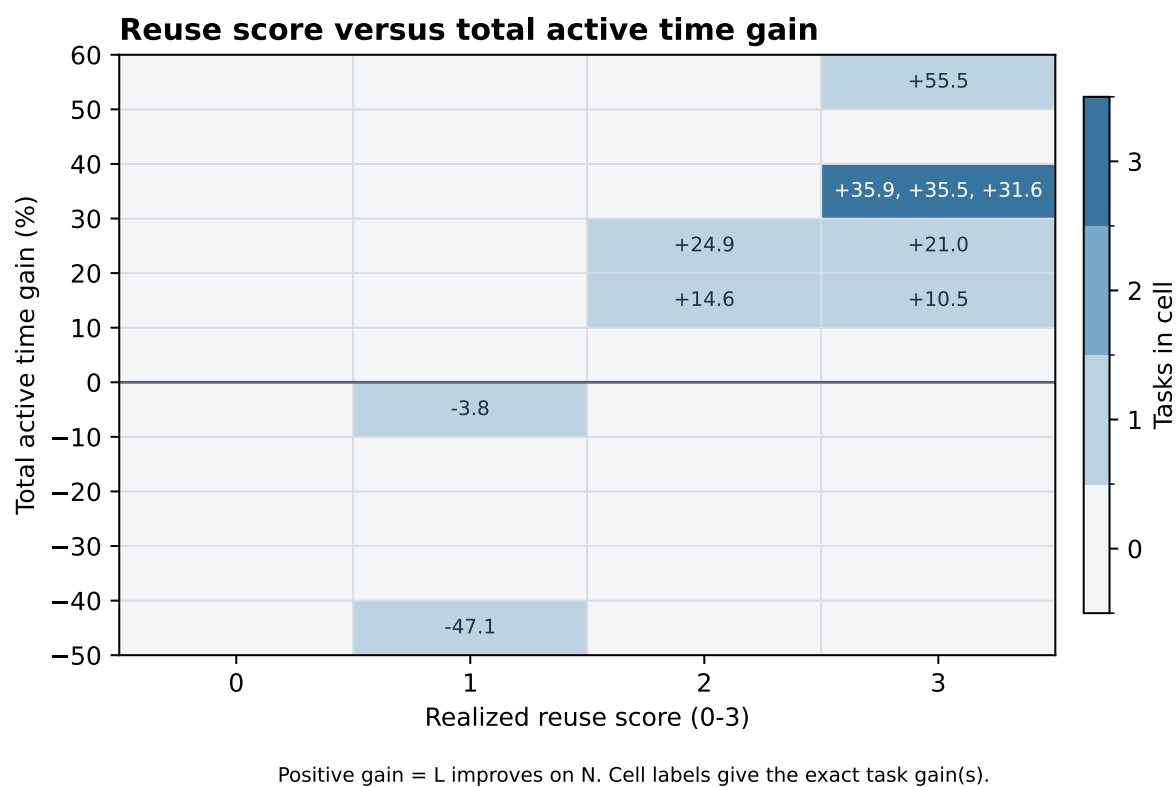


Figure 13: Reuse score versus total contestant-active-time gain, excluding the one-time warm-up. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

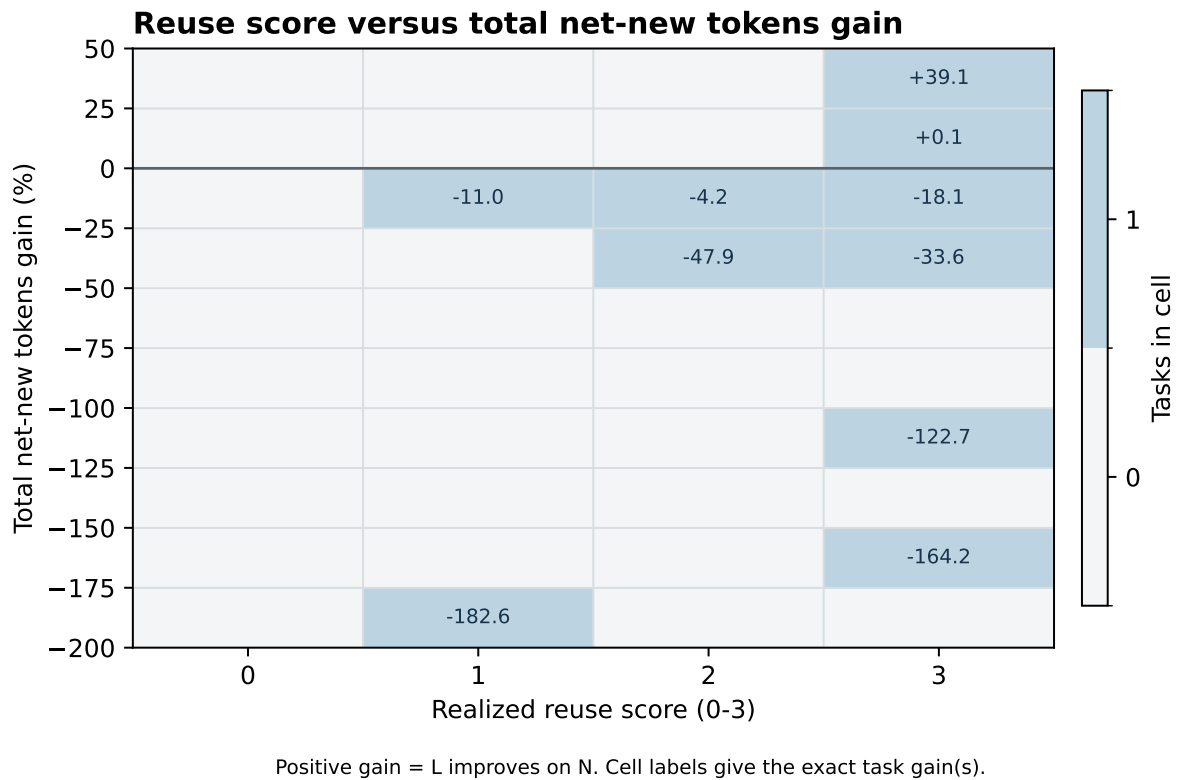


Figure 14: Reuse score versus total net-new-token gain. Negative values mean L used more tokens. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

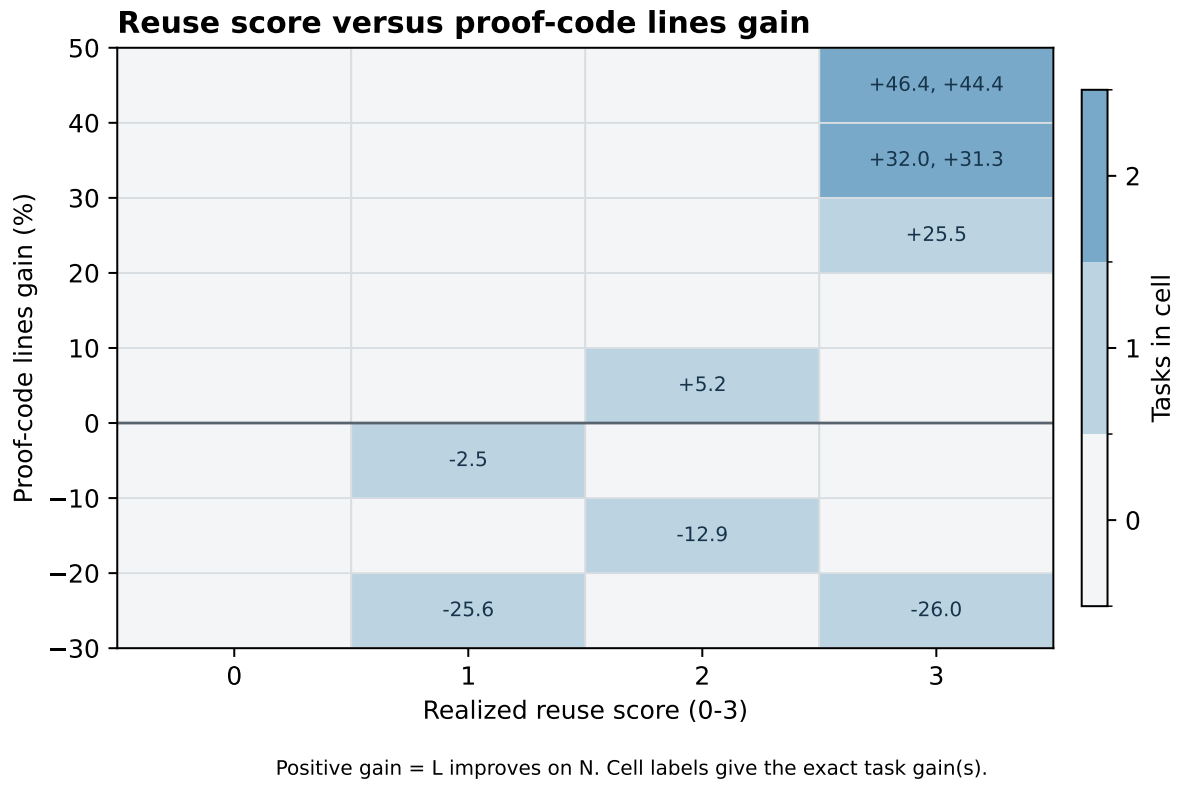


Figure 15: Reuse score versus proof-code-line gain, computed from final kernel-accepted proofs. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

Within these ten, direct proof-name count has Spearman rank correlation $\rho = +0.56$ with total-time gain and $\rho = +0.64$ with proof-line gain. That is a descriptive association only: tasks were selected after outcomes, the number of direct declarations can track proof complexity, and these small-sample coefficients do not identify a causal effect of library coverage. Formalization submissions vary little, so an attempt correlation would be uninformative. Model search appears in archived events, but this trial did not isolate a causal “query overhead” variable.^[1, 15]

The five previously favorable tasks show 34.0% less active time and 37.6% fewer proof lines for L. The five newly screened tasks show 7.2% less active time but 6.4% *more* proof lines. This split is essential: pooled code gains are concentrated in the previously successful group. The cohort chart keeps those identities visible without implying independent sampling.^[1, 4]

5.1 Source-task-specific realized reuse

The three-role index counts (1) task-relevant numeric/FP/gamma/norm *foundation*, (2) computation or rounding *interface*, and (3) proved error/bound or mathematical *analysis*. A role scores one only if an exact witness appears in the final L statement or kernel-accepted proof term; an analysis witness must appear in the proof. Imports, text-only mentions, and transitive-only dependencies score zero. The score is an ordinal 0–3 breadth index, not a count of modules or a proof-line savings estimate. Every task obligation and exact witness is published.^[22, 23, 21]

The rubric was written after outcomes were seen, not by a blind assessor. It is validated against archived statement/proof names and task-packet hashes but cannot turn realized use into prospective library coverage. Six tasks score 3, two score 2, and two score 1. Aggregate active-time gains by score are -9.1% , $+19.3\%$, and $+31.1\%$; proof-line gains are -4.2% , -1.2% , and $+31.4\%$. The pooled score–time rank association is $\rho = +0.72$, while proof-line association is $\rho = +0.58$. For the five newly screened tasks alone, the corresponding coefficients are $+0.63$ and -0.32 . These post-selection, tiny-group statistics must not be read as confirmatory significance tests.^[1, 23, 4]

The two gain–gain heatmaps put formalization against proof time, and total active time against total net-new tokens. Both axes are N-to-L percentage gains; zero lines separate improvement from regression. Cell shade gives the mean reuse score of the task(s) in that bin, and cell text gives each task number and its reuse score (R0–R3). The task-number key is printed inside each figure and exact memberships are in the heatmap ledger. Bins show coarse joint patterns; they do not create additional observations.^[24, 2]

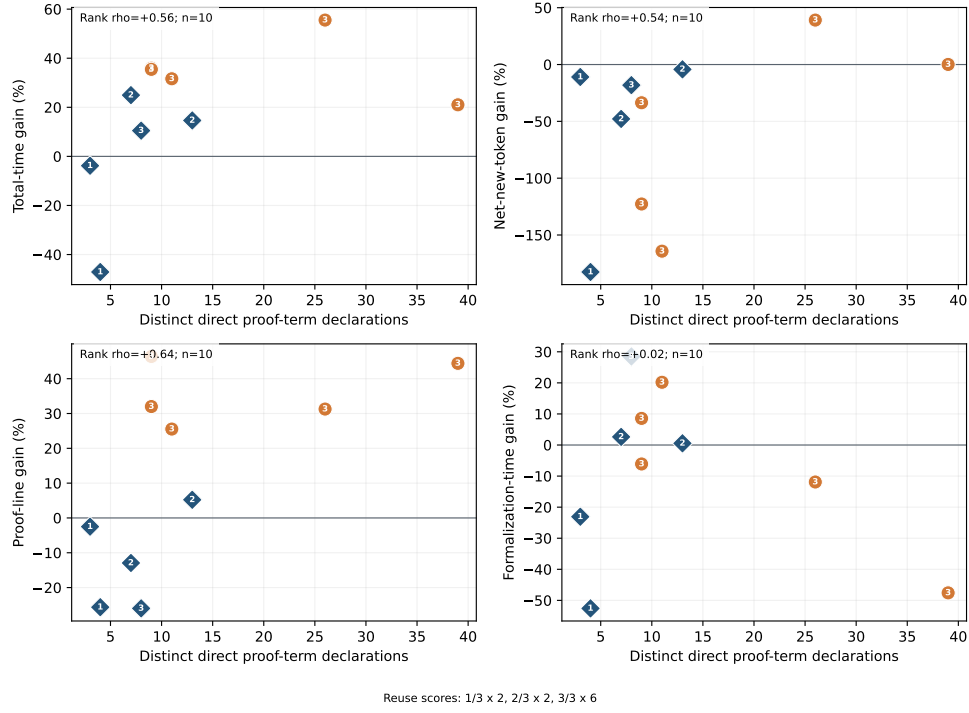


Figure 16: Distinct direct proof-term NumStability names against time, token, proof-line, and formalization-time gains. White numerals are the reuse scores. Rank coefficients are descriptive at $n = 10$. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

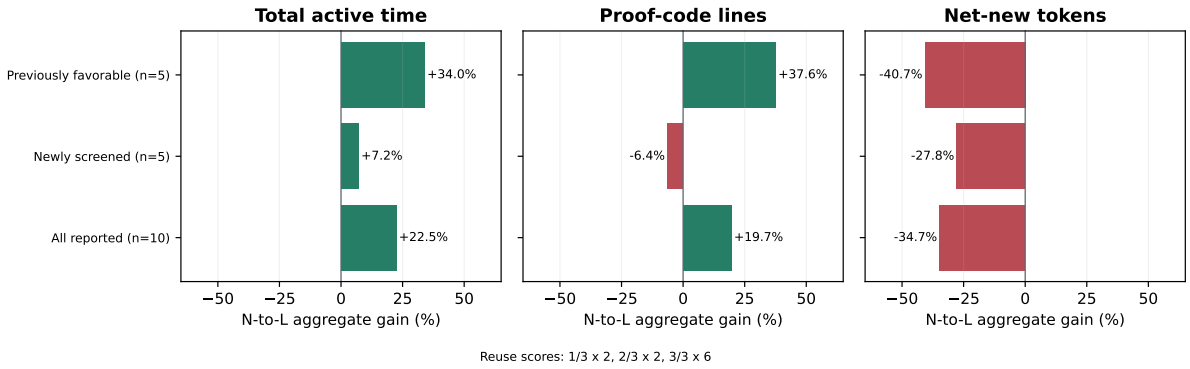


Figure 17: Aggregate N-to-L time, proof-line, and token gains for the five previously favorable tasks, five newly screened tasks, and all ten reported tasks. Each panel includes the reuse-score distribution. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

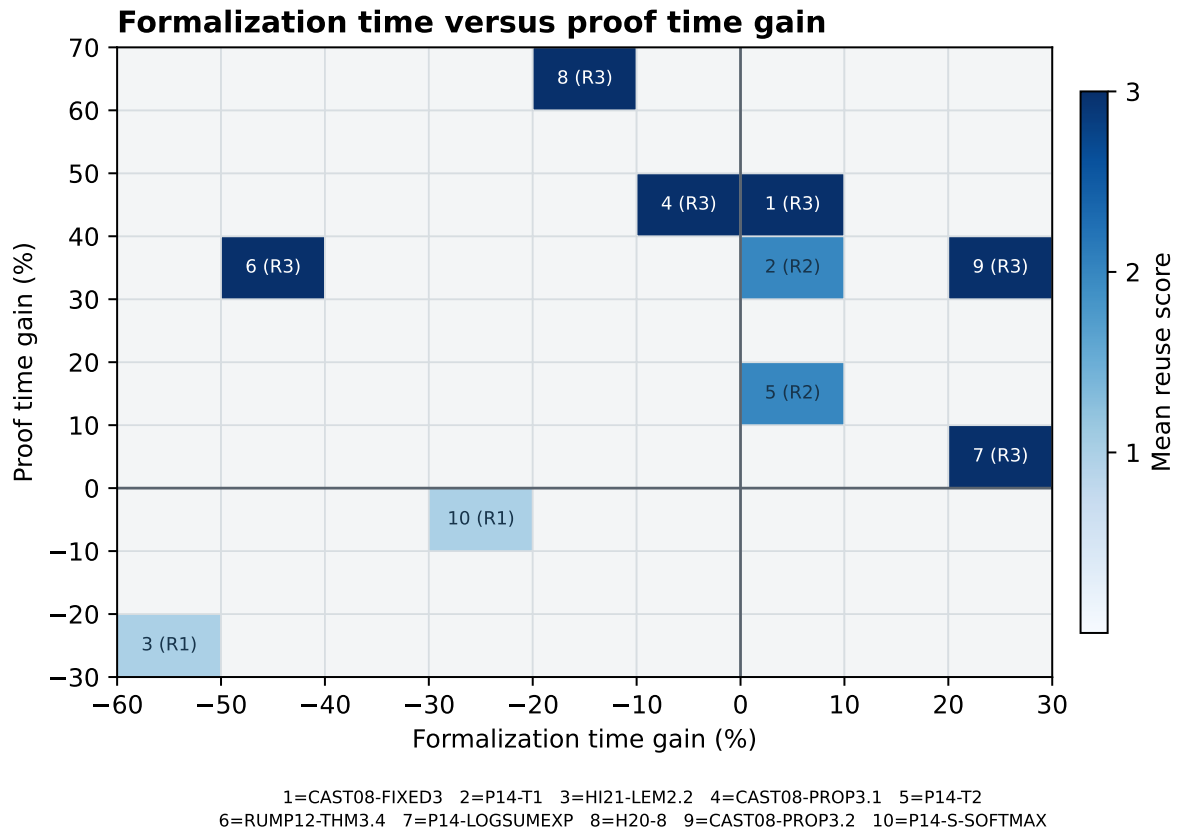


Figure 18: Formalization-time gain versus proof-time gain. Task numbers and reuse scores appear in occupied cells; the upper-right quadrant means L was faster in both phases. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

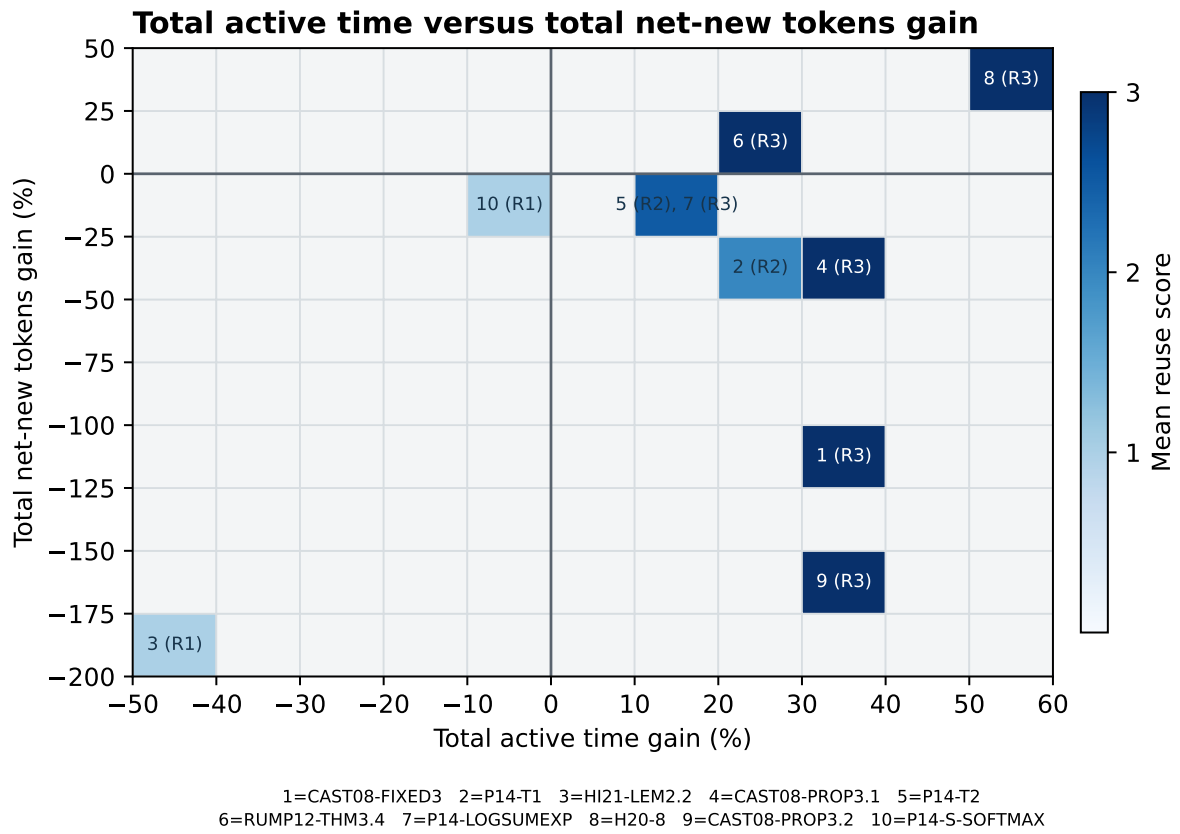


Figure 19: Total active-time gain versus total net-new-token gain. The upper-right quadrant means L was faster and used fewer net-new tokens. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[\[2, 3\]](#)

Concrete examples show reusable material: CAST08-FIXED3 used dot-product and recursive-sum interfaces, reducing proof code from 325 to 221 lines; CAST08-PROP3.1 used summation/error declarations and reduced 373 to 200; H20-8 used least-squares backward-error interfaces and reduced 243 to 167. There are exceptions even at score 3: P14-LOGSUMEXP was faster but its L proof had 257 versus 204 N lines. The score records observed breadth, not what would have happened without the library.^[2, 23, 21]

6 Audit overhead, hardware, and limitations

The faithfulness audit was not a shortcut. Across the reported task records, excluded audit role calls summed to 3,287 seconds and 3.23 million reported tokens for N, and 3,625 seconds and 3.48 million for L. These are sums of role calls, not contestant time or necessarily serial campaign wall time. Compilation, semantic dossiers, and proof validation have separately logged fields. Task formalization and proof stages consumed 27.26 million provider-reported tokens including cached input for N and 55.06 million for L; net-new tokens are the narrower primary comparison.^[1, 2]

Limitations include outcome-informed selection, five previously favorable tasks, four related tasks from one paper, one run per task, no randomization, an audit-schema incident with disclosed recovery, possible equivalent versus stronger accepted statements, and dependence on a changing model service for any future rerun. Warm L knowledge is a measured task-neutral scout, not an actual fine-tuned model. Observed reuse does not isolate the causal value of each declaration or prove that expansion of NumStability alone would remove search, representation, or interface friction.^[4, 2, 8]

The strongest defensible conclusion is specific: in this ten-task, outcome-informed development corpus, NumStability supplied directly used components and L produced shorter and faster complete proofs in aggregate. The highest realized-reuse group had the largest pooled proof-line and time gains, but the new-only code result and token cost temper any general coverage-benefit claim. Independent validation would require a task set and reuse rubric fixed *before* outcomes are observed.^[1, 23]

7 Reproducibility and source trail

This branch provides the report source, plotting program, all generated ten-task tables and figures, task ledger, reuse witnesses, declaration scan, selected paper-source registry, dated selection manifest, frozen protocol, prompts, controller sources, warm-scout record, and measured-pair evidence archives. The script verifies ten exact task identities, paired success, direct score witnesses, and source-input hashes. The repository verifier checks the underlying sealed pair identities and sums. Source papers, authenticated Codex state, and large compiled caches are not repackaged; the source registry provides PDF hashes and task-packet identities. A rerun can follow the pinned protocol but cannot guarantee bit-for-bit identical model behavior.^[25, 1, 5, 15]

Rebuild with the pinned Python requirements, then compile this `report.tex` using `latexmk`. All displayed aggregate rows and task-level appendices are generated from the derived ten-task ledger; figures use unrounded values. The selected evidence is read-only: changing any task, prompt, or library snapshot would require a new benchmark identity rather than an edit to these results.^[3, 25, 2]

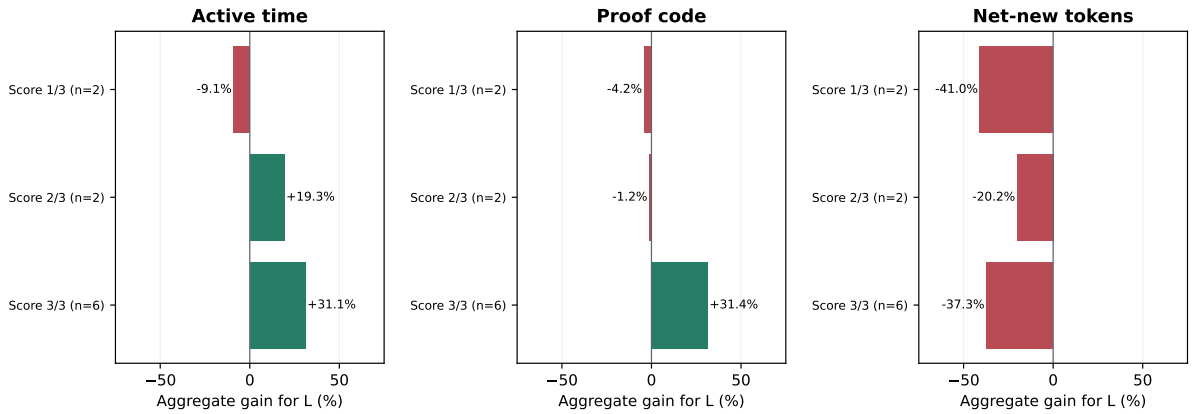


Figure 20: Aggregate gains within displayed score 1, 2, and 3 groups, with sample counts. Code-size improvement is substantial only in score 3; tokens are unfavorable in every score stratum. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

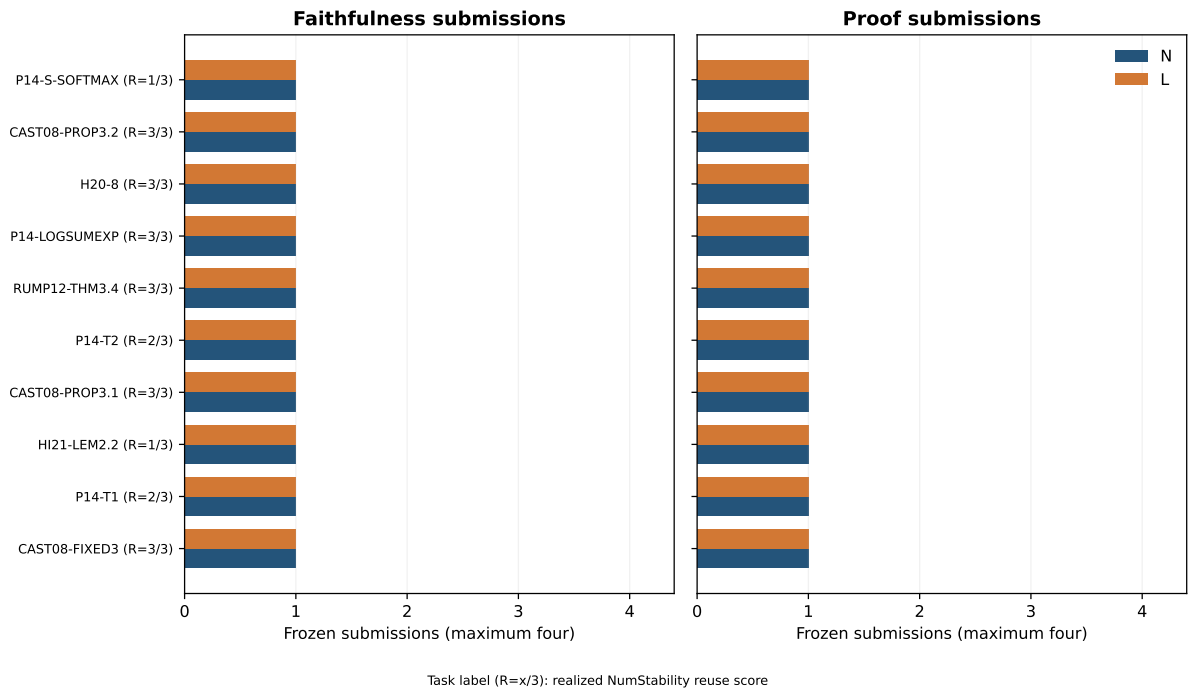


Figure 21: Frozen statement and proof submissions per task and condition, with the reuse score beside every task. Each stage allowed at most four submissions. All task values are derived from the sealed N/L pairs by the reproducible ten-task plotting script.[2, 3]

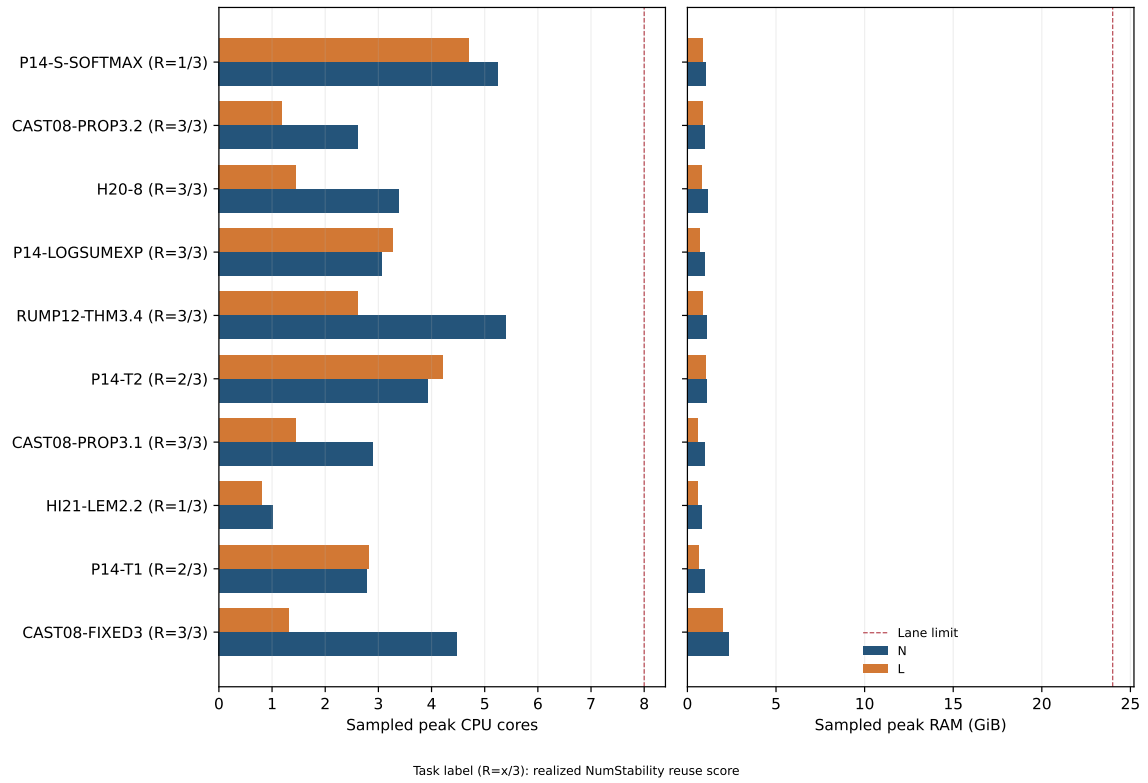


Figure 22: Sampled per-task CPU and RAM peaks against lane caps; brief unobserved spikes are possible. Every task label includes the reuse score. [1, 2]

A Task-level numeric ledger

The tables below contain only the ten reported tasks in their scheduled order. Formalization and proof times are contestant-active seconds; token totals are net-new thousands. SS means substantive statement-declaration count and PT means direct proof-term declaration count. AF/AP are L formalization/proof submission counts. All exact-precision values and both conditions' attempt counts are in the machine-readable ledger. ^[2, 3]

Table 4: Task-level time and net-new token comparison, excluding warm-up. ^[2, 3]

Task	F-N	F-L	P-N	P-L	Total-N	Total-L	Tok-N(k)	Tok-L(k)
CAST08-FIXED3	123	112	481	275	603	387	111	246
P14-T1	176	171	428	282	604	453	87	129
HI21-LEM2.2	106	161	29	37	135	199	40	112
CAST08-PROP3.1	170	180	646	347	816	527	119	159
P14-T2	135	135	590	485	725	619	151	158
RUMP12-THM3.4	156	230	1087	752	1243	982	227	227
P14-LOGSUMEXP	130	93	428	407	558	499	122	144
H20-8	91	102	612	211	703	313	134	82
CAST08-PROP3.2	155	124	464	300	619	423	95	251
P14-SHIFTED-SOFTMAX	148	183	823	826	972	1009	188	208

Table 5: Final code size, declaration reach, and L submission counts. ^[2, 21, 3]

Task	Cluster	Prior	Stmt-N	Stmt-L	Proof-N	Proof-L	SS	PT	AF	AP
CAST08-FIXED3	CAST08	yes	44	39	325	221	2	9	1	1
P14-T1	P14	no	27	29	201	227	2	7	1	1
HI21-LEM2.2	HI21	no	26	36	39	49	4	4	1	1
CAST08-PROP3.1	CAST08	yes	40	39	373	200	1	9	1	1
P14-T2	P14	no	35	37	364	345	0	13	1	1
RUMP12-THM3.4	RUMP12	yes	31	27	599	333	7	39	1	1
P14-LOGSUMEXP	P14	no	28	27	204	257	2	8	1	1
H20-8	H20	yes	25	20	243	167	5	26	1	1
CAST08-PROP3.2	CAST08	yes	37	36	329	245	2	11	1	1
P14-SHIFTED-SOFTMAX	P14	no	40	47	481	493	1	3	1	1

B Realized-reuse task ledger

F, C, and A are foundation, computation/interface, and analysis roles. A “yes” means at least one exact witness was reached in the final L statement or proof. The JSON ledger gives every witness and surface. Positive gains favor L.^[22, 23]

Table 6: Realized-reuse breadth and paired N-to-L gains (percent). [23, 3]

Task	Score	F	C	A	Proof lines	Proof time	Total time	Total tokens
CAST08-FIXED3	3	yes	yes	yes	+32	+43	+36	-123
P14-T1	2	yes	yes	–	-13	+34	+25	-48
HI21-LEM2.2	1	–	yes	–	-26	-27	-47	-183
CAST08-PROP3.1	3	yes	yes	yes	+46	+46	+35	-34
P14-T2	2	yes	–	yes	+5	+18	+15	-4
RUMP12-THM3.4	3	yes	yes	yes	+44	+31	+21	+0
P14-LOGSUMEXP	3	yes	yes	yes	-26	+5	+10	-18
H20-8	3	yes	yes	yes	+31	+66	+56	+39
CAST08-PROP3.2	3	yes	yes	yes	+26	+35	+32	-164
P14-SHIFTED-SOFTMAX	1	yes	–	–	-2	-0	-4	-11

C Direct declaration-usage index

Family labels are descriptive; the JSON declaration scan contains exact elaborated names in statements and proof terms. Imports and text-only mentions are not counted.^[21, 2]

Table 7: Substantive statement and direct proof-term declaration families. ^[21, 3]

Task	Statement family	n	Proof family	n
CAST08-FIXED3	recursive sum, dot product	2	recursive sum, dot product, gamma bound	9
P14-T1	recursive sum	2	recursive sum	7
HI21-LEM2.2	sum tree	4	sum tree	4
CAST08-PROP3.1	recursive sum	1	recursive sum, gamma bound	9
P14-T2	none	0	gamma bound, product error	13
RUMP12-THM3.4	finite FP format	7	finite FP format	39
P14-LOGSUMEXP	recursive sum	2	recursive sum, gamma bound	8
H20-8	least squares	5	least squares	26
CAST08-PROP3.2	recursive sum	2	recursive sum, gamma bound	11
P14-SHIFTED-SOFTMAX	infinity norm	1	infinity norm	3

D Source registry and immutable identities

Every figure and numerical table derives from the ten-task ledger and summary. Method claims are linked to the frozen protocol, prompts, campaign records, hardware snapshots, or proof evidence. The source-selection manifest is retained for auditability; its outcome-informed status is part of the study record.^[1, 4, 25]

Source	Path / identity
Ten-task ledger	<code>ten_task_report/generated/source_task_ledger.json</code> ; its metadata records the sealed-result SHA-256.
Selected source registry	<code>ten_task_report/generated/source_registry.json</code> ; task packets and paper-PDF hashes.
Warm scout	<code>thesis_report/evidence/warm-root.json</code> ; SHA-256 begins 3401426d25ee278c.
Direct declarations	<code>ten_task_report/generated/declaration_scan.json</code> ; exact elaborated names.
Realized reuse	<code>ten_task_report/generated/realized_reuse.json</code> ; task obligations and exact proof/statement witnesses.
Heatmap cells	<code>ten_task_report/generated/heatmap_cells.json</code> ; bin edges, occupied cells, exact tasks and gain values.
Aggregate metrics	<code>ten_task_report/generated/summary.json</code> ; all ten-task sums, score groups, source digests, and overhead.
Selection disclosure	<code>thesis_report/posthoc_10/generated/summary.json</code> ; dated, outcome-informed selection manifest.
Measured pair evidence	Compressed Pilot 34/35 archives under <code>paper_benchmark/thesis_report/evidence/</code> ; archive SHA-256 values in companion README.
Library build	<code>thesis_report/evidence/library-build-record.json</code> ; SHA-256 begins 6f4581380982e51f.
Runtime snapshot	<code>thesis_report/evidence/runtime-snapshot.json</code> ; SHA-256 begins 9fac78511309a591.

References

- [1] This report’s generated ten-task summary, `paper_benchmark/thesis_report/ten_task_report/generated/summary.json`.
- [2] Selected sealed task records, `paper_benchmark/thesis_report/ten_task_report/generated/source_task_ledger.json`; source result digest in metadata.
- [3] Reproducible figure/table script, `paper_benchmark/thesis_report/ten_task_report/make_figures.py`.
- [4] Dated post-outcome task-selection manifest, `paper_benchmark/thesis_report/posthoc_10/generated/summary.json`.
- [5] Selected task packet and paper-source identities, `paper_benchmark/thesis_report/ten_task_report/generated/source_registry.json`.
- [6] Paper/source descriptions and task packets in `paper_benchmark/formalization_benchmark/design33/packets/`.
- [7] Hash-frozen task packets in `paper_benchmark/formalization_benchmark/design33/packets/`; selected hashes in the source registry.
- [8] Proof-required source-first protocol, `paper_benchmark/formalization_benchmark/design27/PROTOCOL.md`.
- [9] Prospective metric contract, `paper_benchmark/formalization_benchmark/design33/METRICS.md`.
- [10] Frozen common formalization prompt and L appendix, `paper_benchmark/formalization_benchmark/design27/prompts/`.
- [11] Strict proof prompt, `paper_benchmark/prospective_proof_no_holes/PROOF_PROMPT.md`.
- [12] Faithfulness-audit role prompts, `paper_benchmark/formalization_benchmark/audit/prompts/`; effective prompt hashes in campaign input records.
- [13] Frozen task-neutral warm-root record, `paper_benchmark/thesis_report/evidence/warm-root.json`.
- [14] Scout prompt and observable transcript, `paper_benchmark/thesis_report/evidence/warm-scout/`.
- [15] Archived measured-pair prompts, candidates, audits, validators, hardware and final Lean files under `paper_benchmark/thesis_report/evidence/`; checksums in the companion README.
- [16] Archived campaign incident and recovery manifests in `paper_benchmark/pilot35/` and the measured-pair evidence archives.
- [17] Frozen campaign runtime and Codex CLI identity, archived campaign inputs in the evidence bundle.
- [18] NumStability full build record, `paper_benchmark/thesis_report/evidence/library-build-record.json`.
- [19] Library snapshot, `paper_benchmark/thesis_report/evidence/library-snapshot.json`.
- [20] Runtime snapshot, `paper_benchmark/thesis_report/evidence/runtime-snapshot.json`.
- [21] This report’s direct declaration ledger, `paper_benchmark/thesis_report/ten_task_report/generated/declaration_scan.json`.
- [22] Source-task-specific, post-outcome three-role rubric, `paper_benchmark/thesis_report/reuse_obligations.json`.
- [23] This report’s exact-witness reuse ledger, `paper_benchmark/thesis_report/ten_task_report/generated/realized_reuse.json`.
- [24] Exact bin edges and task memberships for the seven gain heatmaps, `paper_benchmark/thesis_report/ten_task_report/generated/heatmap_cells.json`.
- [25] Rebuild and source instructions, `paper_benchmark/thesis_report/ten_task_report/README.md`.